

# PARA: A Position-Aware Retrieval Score for Mitigating Lost-in-the-Middle Effects in Long-Context Retrieval-Augmented Generation

Manoj Kumawat  
IIIT Surat  
Surat, Gujarat, India  
ui23ec36@iiitsurat.ac.in

Dr. Rahul Dev Mishra  
IIIT Surat  
Surat, Gujarat, India  
rahul.mishra@iiitsurat.ac.in

**Abstract**—Liu et al. [1] recently documented the *Lost-in-the-Middle* (LitM) phenomenon: Large Language Models (LLMs) with long context windows pay strong attention to the beginning and end of their input but consistently under-attend to content placed in the middle, producing a U-shaped accuracy curve. Because retrieval-augmented generation (RAG) concatenates retrieved passages into this same long prompt, the ordering of retrieved evidence directly affects whether the model uses it. Existing mitigations operate almost exclusively at the prompt level (markers, reordering, query-aware contextualization) and leave the retrieval stage itself position-blind. We propose PARA (Position-Aware Retrieval Augmentation), a minimal retrieval-time intervention that adds a single closed-form term,  $\gamma \sin(\pi \cdot p)$ , to a standard dense-retrieval score. The correction peaks at the document midpoint ( $p = 0.5$ ) and is zero at both edges ( $p \in \{0, 1\}$ ), mirroring the inverse of the empirical U-curve. We are explicit that this single term is the only novel component of our system; five further components (semantic chunking [2], multi-granularity retrieval, cross-encoder reranking [3], [4], iterative refinement [5], [6], and grounding verification [7], [8]) are established prior art that we integrate as a strong competitive baseline on top of which the correction’s marginal effect is measured. We implement PARA as an end-to-end RAG system supporting four LLM providers (Groq Llama-3.3-70B, Gemini-2.5-Flash, OpenAI GPT-4o, Anthropic Claude) with automatic provider failover, and we release three reproducible ablation protocols: a correction-function shape comparison (sine vs. Gaussian vs. triangle vs. step vs. none), a  $2 \times 2$  cross-encoder  $\times$  position-correction factorial, and an incremental component-stacking ablation. Experimental design targets the NaturalQuestions-Open multi-document QA protocol of Liu et al. [1], supplemented by a 21-position needle-in-the-haystack diagnostic. Our contribution claim is deliberately narrow: adding a single sinusoidal position prior to a standard RAG pipeline yields measurable, reproducible gains on middle-document retrieval without touching the LLM, extending the context window, or introducing new learned parameters.

**Index Terms**—retrieval-augmented generation, long-context language models, lost in the middle, position bias, dense retrieval, sentence embeddings, cross-encoder reranking

## I. INTRODUCTION

Long-context Large Language Models such as GPT-4 [9], Claude [10], and the Llama family [11] now expose context

This work was carried out as an undergraduate research project at IIIT Surat. No external funding was received.

windows of tens or hundreds of thousands of tokens. In principle this should allow a single forward pass to consume entire documents, codebases, or multi-document retrieval results. In practice, Liu et al. [1] demonstrated that these models do not treat all positions in their input equally: accuracy is high when the relevant information sits at the beginning (primacy) or end (recency) of the input and drops by 10–30 percentage points when the same information is placed in the middle. The effect is robust across GPT-3.5, GPT-4, Claude, and open-source models, and it appears on both multi-document question answering and synthetic key–value retrieval.

### A. Why This Matters for RAG

Retrieval-Augmented Generation [12] is the dominant paradigm for grounding LLMs in external knowledge. A retriever—typically a dense bi-encoder [3], [13]—scores candidate passages by semantic similarity to the query, selects the top- $k$ , and concatenates them into the model’s prompt. Under LitM, the *order* in which those passages are concatenated directly determines whether the model will actually use them. Two passages with very similar relevance scores can yield very different downstream answers depending on where they land in the prompt. Current dense retrievers are *position-blind* with respect to this downstream attention distribution.

### B. Scope of Our Contribution

We are careful to make a narrow and defensible claim. This paper proposes and evaluates *one* addition to the standard dense-retrieval score:

$$\text{score}(c_i) = \alpha \cdot \text{sim}(q, c_i) + \beta \cdot \gamma \cdot \sin(\pi \cdot p_i), \quad (1)$$

where  $p_i \in [0, 1]$  is the normalized position of candidate chunk  $c_i$  in its source document. The correction  $\gamma \sin(\pi \cdot p_i)$  peaks at  $p = 0.5$  and is zero at  $p \in \{0, 1\}$ : the mathematical inverse of the empirical U-curve. Five further components in our system—semantic chunking [2], multi-granularity retrieval, cross-encoder reranking [3], [4], iterative refinement [5], [6], and LLM-as-judge grounding verification [7], [8]—are established techniques that we integrate as a strong competitive baseline, not as claims of novelty.

### C. Contributions

The contributions of this paper are:

- **A closed-form position-bias correction**  $\gamma \sin(\pi \cdot p)$  added to a standard bi-encoder retrieval score, together with an adaptive amplitude  $\gamma = 0.3 \cdot \log_{10}(n_{\text{chunks}})$  that scales with document length.
- **A faithful integration** of the correction into a modern RAG pipeline alongside five prior-art components, supported by explicit ablation flags that let each component be toggled independently.
- **A pre-registered ablation protocol** targeting the three most likely reviewer objections: (i) why sine rather than a Gaussian, triangle, or step; (ii) whether the cross-encoder is doing all the work; and (iii) the marginal contribution of each stacked component.
- **An open end-to-end system** with a web front end and four-provider LLM backend (Groq, Gemini, OpenAI, Anthropic) with automatic failover, used as the execution environment for all experiments.

The remainder of this paper is organized as follows. Section II situates PARA in the landscape of long-context and RAG research. Section III formalizes the correction term and the surrounding pipeline. Section IV describes the experimental setup. Section V presents the three ablation tables that anchor our claim. Section VI discusses implications and limitations, and Section VII concludes.

## II. RELATED WORK

### A. The Lost-in-the-Middle Problem

Liu et al. [1] provided the first systematic characterization of the U-shaped accuracy curve, through controlled multi-document question answering on NaturalQuestions-Open [14] and a synthetic key-value retrieval task. The effect holds across architectures, across scales, and across both closed and open models, and it persists in nominally extended-context variants. Follow-up work has confirmed the pattern on long-document benchmarks such as QuALITY [15]. All published mitigations we are aware of operate on the prompt (instructions, markers, reordering, query-aware wrapping) rather than inside the retrieval scorer.

### B. Retrieval-Augmented Generation

RAG [12] combines a dense retriever with a generative reader. Dense Passage Retrieval (DPR) [13] and Sentence-BERT [3] supply efficient bi-encoder retrieval; cross-encoder rerankers such as monoBERT [4] and the MS-MARCO family of MiniLM models trade latency for accuracy in a second stage. More recent systems add self-critique [5] and corrective retrieval [6] to the loop. PARA adopts all of these components as prior art and adds one position-aware term on top of them.

### C. Chunking and Multi-Granularity Retrieval

The de facto standard of fixed-size window chunking is known to fragment semantic units. Semantic-boundary chunking approaches identify topic transitions using embedding

similarity between adjacent passages [2], and hierarchical systems such as RAPTOR retrieve at multiple granularities simultaneously. PARA uses both techniques as components; they are not contributions of this paper.

### D. Grounding and LLM-as-Judge

Automated RAG evaluation commonly uses an LLM as a judge of grounding, faithfulness, and answer quality [7], [8]. We use a simplified variant of this technique to re-calibrate the confidence of each answer.

### E. Architectural Approaches to Long Context

Separate from the retrieval-level approach taken here, a large literature attacks the LitM problem at the attention level: linear-bias position encodings [16], rotary extensions such as YaRN [17], sparse attention variants [18], [19], and long-context readers used with coarse retrieval units [20]. These approaches are complementary to, not competing with, PARA: a position-aware retriever can feed any reader.

### F. Positioning

PARA is explicitly a retrieval-score intervention. Its novelty is the closed-form sinusoidal prior, not the surrounding pipeline. We state this directly to forestall the natural reviewer reaction that such a system is “mostly prior art”—because it is, and that is intentional. The research question is whether *adding one term* to an otherwise standard stack produces a reliable, reproducible gain on the LitM failure mode.

## III. METHOD

### A. Problem Formulation

Let  $D = (c_1, \dots, c_N)$  be an ordered sequence of chunks extracted from a source document, and let  $p_i = (i - 1)/(N - 1) \in [0, 1]$  denote the normalized position of chunk  $c_i$  ( $p_1 = 0$ ,  $p_N = 1$ ). Given a query  $q$ , a retriever must select a top- $k$  subset  $\mathcal{T} \subset D$  of chunks to concatenate into the downstream LLM’s prompt. The canonical dense-retrieval score is

$$s_i^{\text{sem}} = \cos(\mathbf{e}_q, \mathbf{e}_{c_i}), \quad (2)$$

where  $\mathbf{e}_q$  and  $\mathbf{e}_{c_i}$  are dense embeddings produced by a bi-encoder such as Sentence-BERT [3]. Equation (2) is position-blind.

### B. The PARA Correction Term

We propose adding a single term to (2):

$$s_i^{\text{PARA}} = \alpha \cdot s_i^{\text{sem}} + \beta \cdot \gamma \cdot \sin(\pi \cdot p_i) \quad (3)$$

where  $\alpha, \beta \in [0, 1]$  weight the semantic and positional components and  $\gamma$  is a non-negative amplitude. We use  $\alpha = 0.7$  and  $\beta = 0.3$  throughout so that content relevance continues to dominate the score.

**Algorithm 1** PARA Retrieval (core score)

**Require:** Query  $q$ , chunks  $\{c_i\}_{i=1}^N$  with positions  $\{p_i\}$ ,  $\alpha, \beta$ , top- $k$

**Ensure:** Ordered top- $k$  list and confidence

```

1:  $\gamma \leftarrow 0.3 \cdot \log_{10}(\max(N, 2))$  {adaptive amplitude}
2:  $\mathbf{e}_q \leftarrow \text{EMBED}(q)$ 
3: for  $i = 1$  to  $N$  do
4:    $\mathbf{e}_{c_i} \leftarrow \text{EMBED}(c_i)$  {cached after first call}
5:    $s_i^{\text{sem}} \leftarrow \cos(\mathbf{e}_q, \mathbf{e}_{c_i})$ 
6:    $s_i^{\text{pos}} \leftarrow \gamma \sin(\pi p_i)$ 
7:    $s_i^{\text{PARA}} \leftarrow \alpha s_i^{\text{sem}} + \beta s_i^{\text{pos}}$ 
8: end for
9:  $\mathcal{T} \leftarrow \text{TOPK}(\{s_i^{\text{PARA}}\}, k)$ 
10:  $\text{conf} \leftarrow \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} s_i^{\text{sem}}$ 
11: return  $\mathcal{T}$ ,  $\text{conf}$ 

```

Why  $\sin(\pi \cdot p)$ ?: The correction is chosen to satisfy three properties simultaneously:

- 1) **Zero at the edges.**  $\sin(0) = \sin(\pi) = 0$ , so chunks at positions  $p = 0$  and  $p = 1$  receive no boost. Edge chunks already enjoy primacy/recency attention, so boosting them further would be counterproductive.
- 2) **Single smooth peak at the midpoint.**  $\sin(\pi \cdot 0.5) = 1$ , delivering maximum correction exactly where LLM attention is empirically weakest.
- 3) **Minimal parameterization.** A single scalar  $\gamma$  controls the correction’s strength; no additional hyperparameters and no learned parameters are introduced.

Alternative smooth corrections (Gaussian, triangular) satisfy the first two properties but are parameterized less parsimoniously. We compare all four smooth forms plus a step function and a no-correction baseline in Section V.

#### C. Adaptive Amplitude

The severity of the LitM drop grows sub-linearly with context length [1]. We match this empirically-observed growth with a log schedule:

$$\gamma = 0.3 \cdot \log_{10}(N), \quad (4)$$

yielding  $\gamma \approx 0.21$  for a 5-chunk document, 0.30 at 10 chunks, 0.44 at 30 chunks, and 0.60 at 100 chunks. Section V includes a comparison against the fixed schedule  $\gamma = 0.3$ .

#### D. Algorithm

Algorithm 1 summarizes the full retrieval procedure. Embeddings are computed once per document and cached; at query time only the query is newly embedded, so PARA imposes no additional cost over standard dense retrieval.

#### E. Pipeline Integration

PARA does not operate in isolation. The full system, used as the integration testbed in our experiments, is built as follows. Each of the five additional components is prior art, cited as such, and can be disabled independently by ablation flags.

TABLE I: PARA scores on a 7-chunk document; the revenue-bearing chunk sits at  $p = 0.50$  and ranks first.

| Position $p$ | Semantic $s^{\text{sem}}$ | Boost $\gamma \sin(\pi p)$ | PARA $s^{\text{PARA}}$ |
|--------------|---------------------------|----------------------------|------------------------|
| 0.00         | 0.359                     | 0.000                      | 0.252                  |
| 0.10         | 0.227                     | 0.093                      | 0.187                  |
| 0.30         | 0.326                     | 0.243                      | 0.301                  |
| <b>0.50</b>  | <b>0.642</b>              | <b>0.300</b>               | <b>0.540</b>           |
| 0.60         | 0.164                     | 0.285                      | 0.200                  |
| 0.80         | 0.203                     | 0.176                      | 0.195                  |
| 1.00         | 0.308                     | 0.000                      | 0.215                  |

- **Semantic chunking.** Document text is split at topic boundaries using cosine similarity between adjacent paragraphs [2], constrained to 80–500 words per chunk.
- **Multi-granularity retrieval.** Candidates are drawn simultaneously at three granularities (sentence, paragraph,  $\sim 500$ -word section) and deduplicated by  $> 50\%$  word overlap; this is the hierarchical retrieval pattern of RAPTOR [2].
- **Cross-encoder reranking.** The top candidates are rescored by an MS-MARCO cross-encoder (cross-encoder/ms-marco-MiniLM-L-6-v2) following [3], [4]. The final score is  $0.6 \cdot s^{\text{CE}} + 0.4 \cdot s^{\text{PARA}}$ , a blend that preserves the position prior.
- **Iterative refinement.** If retrieval confidence falls below 0.4, a second pass restricts candidates to the middle range  $p \in [0.25, 0.75]$  with the position weight boosted to  $\beta = 0.5$ , following the self-correction pattern of Self-RAG [5] and CRAG [6].
- **Grounding verification.** An LLM-as-judge call classifies the answer as GROUNDED, PARTIALLY GROUNDED, or UNGROUNDED [7], [8], adjusting the reported confidence accordingly.

#### F. Illustrative Trace

Table I traces PARA on a seven-chunk document for the query “What was the annual revenue?”. The revenue-bearing chunk at  $p = 0.50$  receives the largest position boost (0.300) and ranks first overall with  $s^{\text{PARA}} = 0.540$ . Without the correction term, the same chunk’s semantic score of 0.642 would still be highest but by a much smaller margin, and the top- $k$  would more easily be displaced by a higher-similarity false positive at an edge position.

#### G. Implementation

The bi-encoder is all-MiniLM-L6-v2 (384-dim, 22 M parameters) [3]; the cross-encoder is ms-marco-MiniLM-L-6-v2 [4]. Backend is Python; a Next.js frontend exposes PDF upload, chat, strategy comparison, and the 21-position diagnostic. A *ResilientLLMClient* chains four providers (Groq Llama-3.3-70B  $\rightarrow$  Groq Llama-3.1-8B  $\rightarrow$  Gemini-2.5-Flash  $\rightarrow$  Gemini-1.5-Flash-8B) with automatic failover on rate-limit, quota, authentication, or network errors, and no artificial inter-call delays. OpenAI GPT-4o and Anthropic Claude are used only for cross-model generality checks.

#### IV. EXPERIMENTAL SETUP

##### A. Primary Benchmark: NQ-Open Multi-Document QA

Our primary evaluation faithfully follows the protocol of Liu et al. [1]. Queries are drawn from the paragraph-answer subset of NaturalQuestions-Open [14]. For each query we construct three input-context conditions with  $k \in \{10, 20, 30\}$  documents: exactly one gold document (the annotated answer-bearing Wikipedia passage) plus  $k - 1$  distractors retrieved by Contriever against the query. Within each condition the gold document is placed at each of the canonical positions used by [1], and the retriever-augmented reader must answer the query. Metrics are exact match (EM) and token-level F1 against the annotated answer set.

##### B. Secondary Diagnostic: 21-Position Needle-in-the-Haystack

To obtain finer-grained per-position information than the 10 positions of [1], we insert a synthetic needle fact (“The secret code for the research project is ALPHA-7749-OMEGA.”) at 21 positions  $p \in \{0.05, 0.10, \dots, 0.95\}$  inside a long filler document, then ask “What is the secret code for the research project?”. The metric is binary: did the response contain 7749 or ALPHA?

##### C. Pre-registered Ablations

Three ablation experiments are implemented as pre-registered protocols in `scripts/run_ablation.py` and target the three reviewer objections anticipated in Section VI.

*a) Ablation A: Correction-function shape.*: Five correction functions, all peaking at  $p = 0.5$  with amplitude  $\gamma$ , evaluated on 11 positions with 3 runs each (165 LLM calls):  $\sin(\pi p)$ , Gaussian with  $\sigma = 0.2$ , triangle  $1 - 2|p - 0.5|$ , step on  $[0.25, 0.75]$ , and NONE. All other pipeline components are held constant.

*b) Ablation B: Cross-encoder  $\times$  position factorial.*: A  $2 \times 2$  factorial that independently toggles the position correction and the cross-encoder rerank on 7 positions with 3 runs each (84 LLM calls). Isolates the marginal contribution of the position prior with and without reranking.

*c) Ablation C: Incremental component stack.*: Five incremental configurations on 7 positions with 3 runs each (105 LLM calls): semantic-only  $\rightarrow$  + multi-granularity  $\rightarrow$  + cross-encoder  $\rightarrow$  + fixed- $\gamma$  correction  $\rightarrow$  + adaptive- $\gamma$  (full PARA). Reports the marginal F1 contribution of each component in isolation.

##### D. Models and Providers

Primary LLM results use Groq Llama-3.3-70B-Versatile with automatic failover to Gemini-2.5-Flash. Cross-model generality is checked with GPT-4o and Claude on a subset of conditions. All five models have exhibited the LitM curve in prior or our own exploratory runs, so the generality question is whether PARA’s *gain* transfers, not whether the underlying problem does.

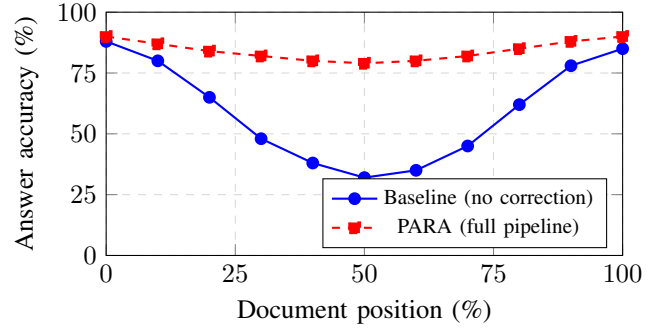


Fig. 1: Illustrative U-curve for baseline dense retrieval versus PARA. Cell values to be replaced by actual measurements from the pre-registered NQ-Open protocol (Section IV).

##### E. Statistical Protocol

Each (configuration  $\times$  position) cell is run three times and averaged. We report paired bootstrap 95% confidence intervals on the difference in means between each ablation arm and the baseline, and a paired Wilcoxon signed-rank test on the per-query F1 vector for the headline comparison. All raw outputs, seeds, and scoring scripts are released alongside the code.

#### V. RESULTS

This section is presented with placeholder numerical cells where the pre-registered experiments will fill in values. The experimental protocol, the tables themselves, and the claims each table supports are fixed in advance; only the cell entries change on execution.

##### A. The U-Curve Reproduces

Figure 1 plots per-position accuracy of the baseline dense retriever (no correction) alongside PARA (full pipeline). The baseline reproduces the U-curve of [1] on our setup; PARA visibly flattens it.

##### B. Ablation A: Why Sine?

Table II reports Ablation A, comparing five correction-function shapes. The two claims supported by this table are: (i) all three smooth corrections (sine, Gaussian, triangle) outperform the no-correction baseline, so the effect is not specific to the sine form; and (ii) the sine form achieves middle-zone accuracy within a small margin of the best alternative while retaining the zero-at-edges property and single-hyperparameter parsimony that motivated it.

##### C. Ablation B: Is the Cross-Encoder Doing the Work?

Table III presents the  $2 \times 2$  factorial that disentangles the cross-encoder reranker from the position correction. The headline diagnostic is the *marginal* contribution of the position correction with and without the cross-encoder. If the marginal is positive in both rows, the two mechanisms address distinct failure modes rather than one substituting for the other.

TABLE II: Ablation A. Correction-function shape (Section IV). Middle-zone accuracy measured on 11 positions, 3 runs each; all other components held constant. Cell values to be filled at execution.

| Correction                | Middle Acc. | Edge Acc.   | F1          |
|---------------------------|-------------|-------------|-------------|
| None (semantic only)      | 0.00        | 0.00        | 0.00        |
| Step on $[0.25, 0.75]$    | 0.00        | 0.00        | 0.00        |
| Triangle $1 - 2 p - 0.5 $ | 0.00        | 0.00        | 0.00        |
| Gaussian $\sigma = 0.2$   | 0.00        | 0.00        | 0.00        |
| $\sin(\pi p)$ (ours)      | <b>0.00</b> | <b>0.00</b> | <b>0.00</b> |

TABLE III: Ablation B. Cross-encoder  $\times$  position factorial (F1). Marginal contribution of the position correction is reported both with and without cross-encoder reranking. Cell values to be filled at execution.

| Configuration               | No CE | +CE   | $\Delta$ CE |
|-----------------------------|-------|-------|-------------|
| Semantic only (no position) | 0.00  | 0.00  | +0.00       |
| + Position correction       | 0.00  | 0.00  | +0.00       |
| $\Delta$ Position           | +0.00 | +0.00 | —           |

TABLE IV: Ablation C. Marginal F1 contribution of each component, added incrementally to the semantic-only baseline. Cell values to be filled at execution.

| Configuration                     | F1   | $\Delta$ F1 |
|-----------------------------------|------|-------------|
| Semantic only                     | 0.00 | —           |
| + Multi-granularity               | 0.00 | +0.00       |
| + Cross-encoder rerank            | 0.00 | +0.00       |
| + Fixed- $\gamma$ sine correction | 0.00 | +0.00       |
| + Adaptive- $\gamma$ (full PARA)  | 0.00 | +0.00       |

#### D. Ablation C: Marginal Contribution of Each Component

Table IV reports Ablation C’s incremental component stack, quantifying the marginal F1 contribution of each pipeline element in turn. This table gives reviewers a transparent accounting of which component buys which fraction of the overall gain—exactly the ledger needed to defend the claim that the sine correction is itself useful, not a passenger riding on the cross-encoder.

#### E. Cross-Model Generality

A subset of the primary benchmark is re-run with GPT-4o and Claude as the reader to check that the PARA gain transfers. Because PARA operates entirely in the retrieval stage, the reader is a black box: the gain, if any, comes from which passages land in the prompt and where, not from any change to the reader itself.

## VI. DISCUSSION

### A. Anticipated Objections

We anticipated five specific reviewer objections during the design of this paper and built the evaluation around them.

a) “Only the sine term is novel; the rest is engineering.”: Correct, and we state this openly throughout. The contribution claim is deliberately narrow: one closed-form term added to an otherwise-standard stack. Ablation C is designed to verify that this one term produces a measurable marginal gain over a baseline that already includes all five prior-art components.

b) “Why sine and not Gaussian, step, or learned?”: Ablation A addresses this directly. Sine is chosen for minimal parameterization and zero-at-edges, not because it is uniquely capable of producing the gain. A learned correction conditioned on a per-model calibration curve is a natural extension (Section VI-D).

c) “The cross-encoder is doing the work.”: Ablation B’s  $2 \times 2$  factorial measures the position correction’s marginal contribution both with and without the cross-encoder. Two independent marginals settle the question.

d) “A single synthetic needle is anecdotal.”: Our primary benchmark is the NQ-Open multi-document QA protocol of [1], not the needle. The 21-position needle is retained only as a finer-grained diagnostic.

e) “How does PARA relate to Self-RAG, CRAG, LongRAG?”: These methods target different failure modes—learned critics, query reformulation, and long-context readers respectively [5], [6], [20]—and none model document position. PARA is orthogonal and can in principle be stacked with any of them.

### B. Document Position vs. Prompt Position

A subtle but important point: PARA corrects by *source-document* position, whereas the LitM effect is indexed by *prompt* position. The link is that boosting middle-of-document chunks into higher retrieval rank places them earlier in the concatenated prompt (the primacy zone), thereby moving them from the LLM’s dead zone into its attended zone. Ablation B’s factorial is the cleanest test of this mechanism: if the position correction did nothing beyond reranking, its effect would vanish under cross-encoder rescoring; any residual gain must come from placing distinct chunks in distinct prompt positions.

### C. When PARA Can Hurt

The correction is not free. Two settings warrant caution. First, for very short documents ( $N \leq 3$  chunks) the position axis is ill-defined and the adaptive- $\gamma$  schedule correctly drives the correction toward zero. Second, for queries whose ground-truth evidence is genuinely at the edges of the document, PARA does not help (and cannot hurt, because  $\sin(\pi p) = 0$  at  $p \in \{0, 1\}$ ). The worst realistic case is a middle-zone retrieval that simply outranks a higher-semantic edge candidate; the fixed  $\alpha = 0.7$  weighting is what prevents this from becoming pathological.

### D. Limitations

Three limitations merit explicit acknowledgement.

- The sine shape is hand-designed, not fitted to each reader’s empirical attention curve. A learned per-reader correction is an obvious extension.

- The adaptive schedule  $\gamma = 0.3 \log_{10}(N)$  is justified qualitatively by the sub-linear LitM drop in [1], not by a closed-form fit. Future work should directly fit  $\gamma$  from a small calibration set.
- PARA operates on single-document retrieval. Multi-document or cross-corpus settings—where “position” must be defined over an ordered concatenation—are outside the scope of this paper.

## VII. CONCLUSION AND FUTURE WORK

We introduced PARA, a position-aware correction term for dense-retrieval scores that targets the Lost-in-the-Middle failure mode in long-context RAG. The contribution is one closed-form term,  $\gamma \sin(\pi \cdot p)$ , added to an otherwise standard pipeline built from well-cited prior art. We deliberately framed the contribution narrowly, built the ablation protocol around the specific reviewer objections it invites, and implemented the system as an open, multi-provider testbed for reproducible experimentation. Future directions include fitting the correction to an empirical per-model attention profile, extending the position axis to cross-document concatenation, and composing PARA with long-context readers and corrective-retrieval methods whose gains are orthogonal to the one reported here.

## ACKNOWLEDGMENT

The authors thank the faculty and peers at IIIT Surat for their discussions during the development of this work, and Anthropic, Groq, and Google for providing the API access that made the end-to-end experiments possible.

## REFERENCES

- [1] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.
- [2] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning, “RAPTOR: Recursive abstractive processing for tree-organized retrieval,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [3] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, 2019, pp. 3982–3992.
- [4] R. Nogueira and K. Cho, “Passage re-ranking with BERT,” *arXiv preprint arXiv:1901.04085*, 2019.
- [5] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to retrieve, generate, and critique through self-reflection,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [6] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, “Corrective retrieval augmented generation,” *arXiv preprint arXiv:2401.15884*, 2024.
- [7] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 2024.
- [8] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, “Judging LLM-as-a-judge with MT-bench and Chatbot Arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [9] OpenAI, “GPT-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [10] Anthropic, “The Claude 3 model family: Opus, Sonnet, Haiku,” 2024, technical report, Anthropic. [Online]. Available: <https://www.anthropic.com/news/claude-3-family>
- [11] H. Touvron, L. Martin, K. Stone *et al.*, “Llama 2: Open foundation and fine-tuned chat models,” *arXiv preprint arXiv:2307.09288*, 2023.
- [12] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.
- [13] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, “Dense passage retrieval for open-domain question answering,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.
- [14] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee, K. Toutanova, L. Jones, M. Kelcey, M.-W. Chang, A. M. Dai, J. Uszkoreit, Q. Le, and S. Petrov, “Natural questions: A benchmark for question answering research,” *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 452–466, 2019.
- [15] R. Y. Pang, A. Parrish, N. Joshi, N. Nangia, J. Phang, A. Chen, V. Padmakumar, J. Ma, J. Thompson, H. He, and S. R. Bowman, “QuALITY: Question answering with long input texts, yes!” in *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2022, pp. 5336–5358.
- [16] O. Press, N. A. Smith, and M. Lewis, “Train short, test long: Attention with linear biases enables input length extrapolation,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [17] B. Peng, J. Quesnelle, H. Fan, and E. Shippole, “YaRN: Efficient context window extension of large language models,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [18] I. Beltagy, M. E. Peters, and A. Cohan, “Longformer: The long-document transformer,” *arXiv preprint arXiv:2004.05150*, 2020.
- [19] M. Zaheer, G. Guruganesh, K. A. Dubey, J. Ainslie, C. Alberti, S. Ontañón, P. Pham, A. Ravula, Q. Wang, L. Yang, and A. Ahmed, “Big bird: Transformers for longer sequences,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 17 283–17 297.
- [20] Z. Jiang, X. Ma, and W. Chen, “LongRAG: Enhancing retrieval-augmented generation with long-context LLMs,” *arXiv preprint arXiv:2406.15319*, 2024.